Contents

Chapter 1

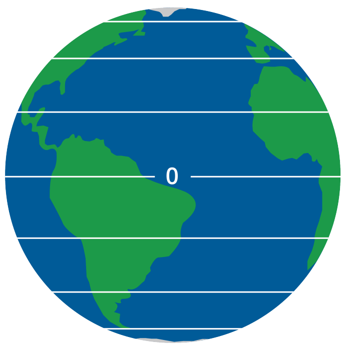
Longitude and Latitude

1.1 Longitude and Latitude

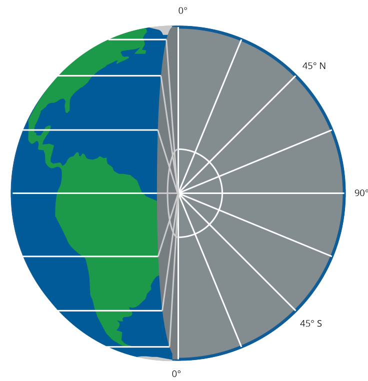
The Earth can be represented as a sphere, and the position of a point on its surface can be described using two coordinates: latitude and longitude.

Figure 1.1: A diagram of latitude and longitude.

Latitude is a measure of a point's distance north or south of the equator, expressed in degrees. It ranges from $-90°$ at the South Pole to $+90°$ at the North Pole, with $0°$ representing the Equator. (See figures 1.2a and 1.2b)



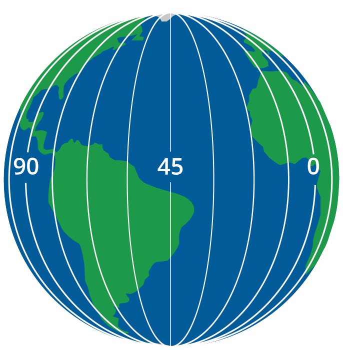
(a) Latitude drawn on the earth.



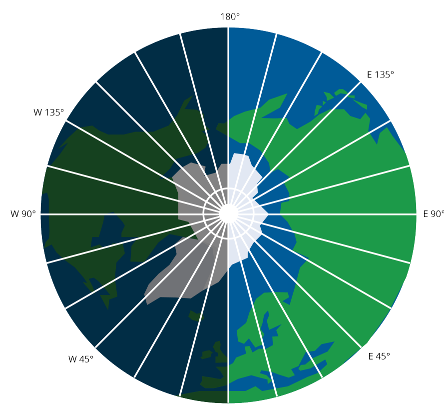
(b) A cross-section of the earth showing latitude.

Figure 1.2: Latitude illustrations

Longitude, on the other hand, measures a point's distance east or west of the Prime Meridian (which passes through Greenwich, England). It ranges from $-180°$ to $+180°$, with the Prime Meridian represented as $0°$. (See figures 1.3a and 1.3b)



(a) Longitudinal lines drawn on the earth.



(b) A cross-section of the earth showing longitude.

Figure 1.3: Longitude illustrations.

The coordinates follow a system of latitude and then longitude (in the majority of contexts). Here is the Longitude and Latitude of four different large cities:

- **London, England**: 51° N 0° W
- **New York City, New York, USA**: 40° N 74° W

- **Sydney, New South Whales, Australia**: 33° S 150° E
- **Rio de Janeiro, Brazil**: 22° S 43° W

These coordinates may be broken down further into minute and second components as well, for further geometric accuracy. Each degree (°) is divided into 60 minutes (′). Each minute (′) is divided into 60 seconds ("). Just like in time. Note some do not

- **London, England**: 51° 30‘ 26″ N 0° 7‘ 39″ W
- **New York City, New York, USA**: 40° 42‘ 46″ N 74° 0‘ 22″ W
- **Sydney, New South Whales, Australia**: 33° 52‘ 0″ S 150° 12‘ 0″ E
- **Rio de Janeiro, Brazil**: 22° 54‘ 0″ S 43° 12‘ 0″ W

1.2 Nautical Mile

A nautical mile is a unit of measurement used primarily in aviation and maritime contexts. It is based on the circumference of the Earth, and is defined as one minute (1/60°) of latitude; in other words, **one minute of latitude is equal to 1 nautical mile.** This makes it directly related to the Earth's geometry, unlike a kilometer or a mile, which are arbitrary in nature. The exact value of a nautical mile can vary slightly depending on which type of latitude you use (e.g., geodetic, geocentric, etc.), but for practical purposes, it is often approximated as 1.852 kilometers or 1.15078 statute miles.

1.3 Haversine Formula

The *haversine formula* is an important equation in navigation for giving great-circle distances between two points on a sphere from their longitudes and latitudes. It is especially useful when it comes to calculating distances between points on the surface of the Earth, which we represent as a sphere for simplicity. See Figure 1.4

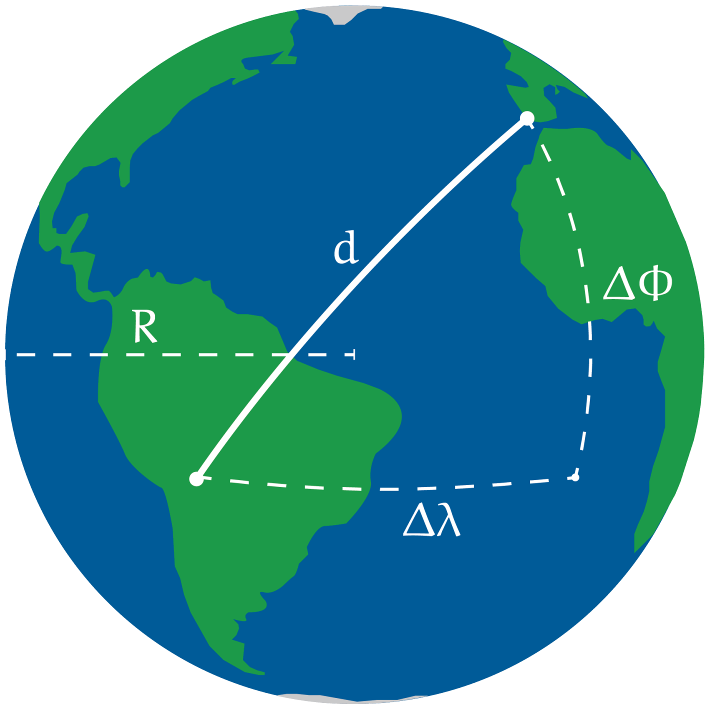


Figure 1.4: A diagram of the haversine formula.

In its basic form, the haversine formula is as follows:

The angle found using haversine haversine$(\theta) = sin^2\left(\frac{\theta}{2}\right)$:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right)$$

Compute c which determines the angular distance using $atan2$, a computer science tangential function with 2 arguments equivalent to $2\arcsin(\sqrt{a})$[1]:

$$c = 2 \cdot \text{atan2}\left(\sqrt{a}, \sqrt{1-a}\right)$$

Find d, the true distance along the sphere:

[1]Note that this function is not available on calculators, but commonly used in computer science programs. You may use the singular argument equivalent in calculations.

$$d = R \cdot c$$

In one line, this is:

$$d = 2R \cdot \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right)}\right)$$

Here, ϕ_1 and ϕ_2 represent the latitudes of the two points (in radians), $\Delta\phi$ and $\Delta\lambda$ represent the differences in latitude and longitude (also in radians), and R is the radius of the Earth (using whatever unit of distance you desire). The result, d, is the distance between the two points along the surface of the sphere. Notice that d is equal to the formula for the *arc length* of a sector. A sphere is just a bunch of circles with radius r, so it follows that we can use formulas for circles in our spherical calculations.

This is why planes fly arc-shaped paths rather than straight paths from points $A \rightarrow B$. Planes follow the shortest path between two points on a sphere, called a *great-circle route*. On a map, like plane flight-progress maps or the Mercator projection, the straight path looks very arc shaped, but this is just a geographical illusion. This straight distance can be calculated using the haversine formula, ultimately minimizing distance and fuel using the 'as-the-crow-flies' path.

Chapter 2

Sampling Data

2.1 Planning a Study

Every day, decisions get made based on data. A city council votes on a new transit route based on ridership surveys. A school principal adjusts the lunch schedule based on student feedback. A pharmaceutical company decides whether to release a new drug based on clinical trial results. In every case, the quality of the decision depends entirely on the quality of the data behind it.

To collect data well, you need a good plan! That plan must specify what question you are trying to answer, how you will gather information, and who you will gather this information from.

2.1.1 Vocabulary

Five terms form the foundation of everything that follows. Learn to distinguish them from each other, because you will probably be tested on this.

population The entire group of individuals or items that you want to learn about. The population is defined by the question you are asking.

frame Also called a sampling frame. A list of all the members of the population from which a sample will be drawn. The frame is usually close to the population, but the two are not always identical. A school's enrollment database, a city's voter registration list, and a company's customer records are all examples of frames.

sample The part of the population that is actually studied. You collect data from the sample and use it to draw conclusions about the population.

sample survey The process of collecting information from a sample. The information gathered is then used to make inferences about population parameters.

census The process of collecting information from every member of the population rather than just a sample. A census gives complete information, but it comes at a cost.

The relationship between these terms is important. The population is the full group you care about. The frame is your working list of that group. The sample is the slice of the frame you actually study. A census is what happens when the sample and the population are the same thing or equal in number.

Exercise 1 Identifying the Population and Sample

A city's parks department wants to understand how often residents use public parks. Staff randomly select 200 households from the city's water billing records and conduct phone interviews.

1. What is the population?
2. What is the frame?
3. What is the sample?
4. Is this a census? Explain.

Working Space

Answer on Page 21

2.1.2 Why Not Always Use a Census?

A census tells you about everyone in the population. That sounds great, but it is not always realistic. If the group is small, like a football team, you can measure everyone.

Most of the time, a census is too difficult because:

- The population is too large. A company that wants to know what all households watch cannot contact every single one. - The measurement destroys the item. If testing a battery kills it, checking every battery would waste them all. - The information gets old too fast. If a company waits months to ask everyone, the results may no longer match what people think.

Exercise 2 Census or Sample?

Working Space

For each situation, decide whether a census is feasible or whether a sample should be used. Briefly explain your reasoning.

Situation	Census or Sample? Explain
A school librarian wants to know which of her 8 student volunteers prefers morning or afternoon shifts.	
A battery manufacturer wants to test the lifespan of batteries by running them until they die.	
A government agency wants to estimate the average commute time of workers across a country of 40 million employed adults.	
A restaurant owner wants to know the dietary restrictions of the 22 guests attending a private event.	

Answer on Page 21

In most studies, sampling is the only practical option, and the quality of the conclusions you draw depends entirely on how well the sample was chosen.

2.2 Planning and Conducting Surveys

Not all samples are fair. How you pick people can make the group represent the population well, or it can leave out important parts.

2.2.1 Biased Sampling

A sample is *biased* if it favors some people or answers over others. Bias does not go away just because the sample is bigger.

These three common methods usually make biased samples.

Judgmental sampling is when one person decides who is included based on their own idea of what looks representative. A school inspector who visits only classrooms she already thinks are good is using judgmental sampling. The choice is based on their opinion.

A *convenience sample* uses whoever or whatever is easiest to reach. A journalist who interviews only people at the bus stop outside his office is using a convenience sample. Those people might not represent all commuters.

A *voluntary response sample* happens when people choose whether to answer. An online poll on a news site will attract mostly visitors with strong opinions. People who do not care much usually do not respond.

All three skip chance. Sampling methods based on random selection give everyone a fair shot and usually make more trustworthy results.

2.3 Simple Random Sampling

A *simple random sample* (SRS) is a sample chosen so every group of the same size has the same chance of being picked.

There are two ways to do it.

With *sampling with replacement*, you put someone back in the pool after choosing them. They could be picked again.

With *sampling without replacement*, you do not put them back. Each person can only be chosen once.

To get an SRS, you need a real random method.

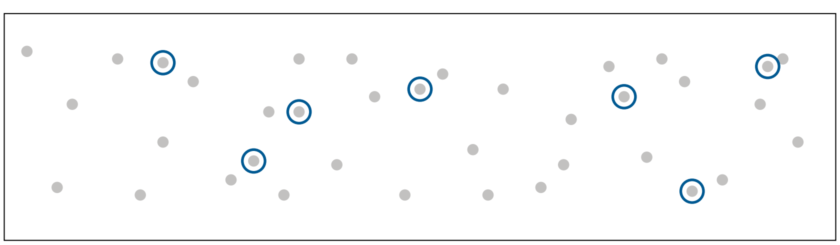


Figure 2.1: In simple random sampling, every individual in the population has an equal chance of being selected, and there is no grouping or pattern to who is chosen.

- Assign every member of the population a number. Use a random number table, entering at any random point and reading numbers sequentially. Select members whose numbers appear, skipping values outside your range and any repeats, until you have the sample size you need.

- Use a calculator. On a TI-84, `MATH` → `PRB` → `5:randInt(` generates random integers. The command `randInt(1,500,30)` produces 30 random integers between 1 and 500. Ignore repeats.
- Use a physical randomization device. Write each member's identifier on an identical slip of paper, mix thoroughly in a container, and draw without looking until the sample is complete.

The essential requirement in every case is that chance dictates who is selected, rather than convenience or judgment. This is what makes the sample fair and the results trustworthy.

Exercise 3 Selecting an SRS

A regional transit authority wants to survey 25 of its 300 bus drivers about workplace safety conditions.

1. Describe a complete procedure for selecting an SRS of 25 drivers using a random number table.
2. A supervisor suggests selecting the 25 drivers who are easiest to reach during the next shift change. What type of sample is this, and what bias might it introduce?
3. A colleague suggests selecting 5 drivers at random from each of the authority's 5 depots. Is this an SRS? Explain.

Working Space

Answer on Page 21

Sampling Real Data from the Web

In this exercise you will load real world population data published by the World Bank, treat all 265 countries and territories as your population, draw a simple random sample, and compare your sample mean to the true population mean.

1. Import the necessary libraries.

```python
import pandas as pd
```

2. Load the dataset directly from the web and filter to the year 2020.

```python
url = "https://raw.githubusercontent.com/datasets/population/main/data/population.csv"
df = pd.read_csv(url)

pop2020 = df[df["Year"] == 2020][["Country Name", "Value"]].reset_index(drop=True)
print(pop2020.head())
print(f"Total entries: {len(pop2020)}")
```

3. Calculate the true population mean. This is the parameter you are trying to estimate.

```python
true_mean = pop2020["Value"].mean()
print(f"True population mean: {true_mean:,.0f}")
```

4. Draw an SRS of 20 countries and compute the sample mean.

```python
sample = pop2020.sample(n=20)
sample_mean = sample["Value"].mean()
print(f"Sample mean (n=20): {sample_mean:,.0f}")
```

5. Run step (d) five times and record the sample mean each time. How close is each to the true mean? Are they all the same?
6. Change the sample size to 80 and repeat step (e). How do the results compare? What does this tell you about the relationship between sample size and sampling error?

2.3.1 Other Methods of Random Sampling

Simple random sampling is not always the most practical or most efficient approach. Three other probabilistic methods appear regularly on the AP exam.

In *systematic sampling*, you list the population, choose a random start, and then pick every kth person. For example, to sample 40 students from 800, start at a random number between 1 and 20, then take every 20th student. This is simple when you have a full list.

In *stratified random sampling*, the population is split into groups called strata. Each group should be similar inside, like students in the same grade or patients in the same ward. Then you take a random sample from each group. This makes sure every part of the population is represented. This method of random sampling is especially useful in medical studies, market research, political polling, educational testing, etc, where you want to make sure you have enough people from each group to compare them.

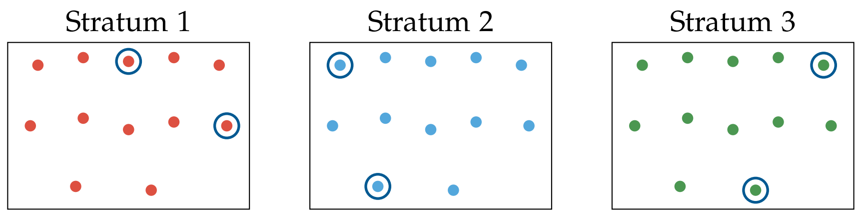


Figure 2.2: Each stratum is a single color, since members within a stratum share a characteristic. A small random sample (circled) is drawn from each stratum.

In *cluster sampling*, the population is split into groups called clusters. Each cluster should be like a mini-version of the whole population. Then you randomly pick some clusters and study everyone inside them. This works well when people are spread out over a wide area and it may be difficult or expensive to reach everyone. For example, a company that wants to survey its customers across the country might randomly select 10 cities and then survey all customers in those cities.

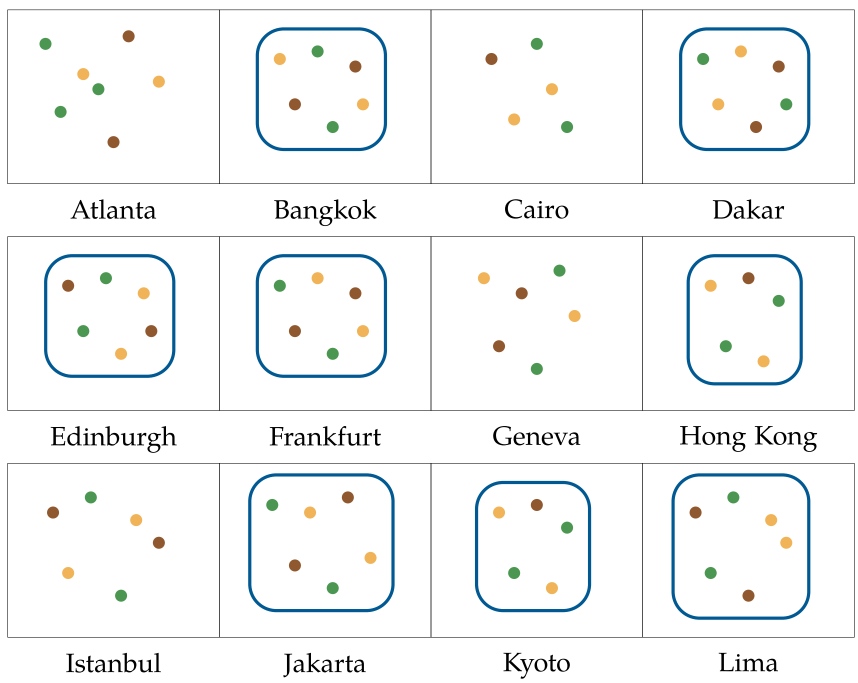


Figure 2.3: Each dot represents a customer and each color represents a different customer segment. Atlanta, Cairo, Geneva, and Istanbul were not selected. The other 8 cities were randomly selected as clusters, and every customer within each selected city becomes part of the sample.

It is important to understand the difference between strata and clusters. *Strata* are groups that are different from each other but similar within, and you sample from all of them. *Clusters* are groups that are similar to the whole population, and you sample only some of them but take everyone inside those.

In *quota sampling*, a non-probability sampling method where researchers select participants until predefined quotas for certain groups (ie: age, gender, ethnicity) are filled. The sample is structured to reflect specific population characteristics, but participants are not chosen randomly. For example, if a population is 60% female and 40% male, a researcher surveys people until they reach those proportions.

In *convenience sampling*, a non-probability sampling method where participants are selected because they are easy to access or available. For example, surveying students who happen to be in a campus cafeteria as you are eating lunch.

Bonus method: *Snowball sampling* is a non-probabilistic method used when the population of interest is difficult or impossible to reach through conventional sampling. The researcher identifies a small number of initial participants who meet the study criteria, and then asks those participants to refer others they know who also qualify. Each wave of referrals generates the next, like a snowball rolling downhill and growing as it goes. It is not a commonly tested method on the AP exam, but it is worth knowing as an example of a non-probabilistic sampling technique that can be useful in certain contexts, such as studying hidden or hard-to-reach populations (e.g., people experiencing homelessness, members of stigmatized groups, or individuals with rare diseases).

Exercise 4 Identifying Sampling Methods

Working Space

Identify the sampling method used in each scenario: SRS, systematic, stratified, or cluster.

Scenario	Method
A supermarket chain selects 10 of its 80 stores at random and surveys every employee in those stores.	
A university randomly selects 50 students from each of its four faculties to participate in a wellbeing survey.	
A water utility selects a random starting meter on its numbered list and then tests every 25th meter for contamination.	
A film festival puts all 900 registered attendee names in a database and uses software to randomly select 90.	

Answer on Page 22

Comparing Sampling Methods in Python

In this exercise you will apply three sampling methods to the same dataset and compare what each one produces. The dataset contains 30 students across three grade levels with their test scores.

Setup: Create the dataset.

```
import pandas as pd
import numpy as np

np.random.seed(42)

data = {
    "student_id": range(1, 31),
    "grade": ["9th"]*10 + ["10th"]*10 + ["11th"]*10,
    "score": (
        list(np.random.randint(60, 80, 10)) +
        list(np.random.randint(70, 90, 10)) +
        list(np.random.randint(80, 100, 10))
    )
```

```
}

df = pd.DataFrame(data)
print(df)
```

Compute the true population mean.

```
true_mean = df["score"].mean()
print(f"True mean score: {true_mean:.1f}")
```

1. **Simple Random Sample.** Draw an SRS of 9 students.

```
srs = df.sample(n=9)
print(f"SRS mean: {srs['score'].mean():.1f}")
print(srs[["student_id", "grade", "score"]])
```

   Run this three times. What grades tend to appear? Does the SRS always represent all three grades?

2. **Stratified Random Sample.** Draw 3 students from each grade level.

```
stratified = df.groupby("grade", group_keys=False).apply(
    lambda x: x.sample(n=3)
)
print(f"Stratified mean: {stratified['score'].mean():.1f}")
print(stratified[["student_id", "grade", "score"]])
```

   Run this three times. How does the sample mean compare to the true mean? Is it more or less stable than the SRS?

3. **Cluster Sample.** Randomly select one entire grade level as the cluster.

```
chosen_grade = pd.Series(df["grade"].unique()).sample(n=1).values[0]
cluster = df[df["grade"] == chosen_grade]
print(f"Cluster selected: {chosen_grade}")
print(f"Cluster mean: {cluster['score'].mean():.1f}")
```

   Run this three times. How much does the cluster mean vary? Why?

4. Fill in the table with your results.

Method	**Run 1**	**Run 2**	**Run 3**	**Most stable?**
SRS				
Stratified				
Cluster				
True mean	79.7			

5. Why does the stratified sample tend to produce a more stable estimate than the SRS? Why does the cluster sample vary the most?

2.4 Bias in Surveys

Even a good random sample can still give bad results if the survey is set up badly. Bias can happen at many steps, further down the line from when you pick people.

Sampling error is the normal difference between a sample and the population. Different samples can give different answers just by chance. This is not the same as bias. Sampling error gets smaller with a bigger sample.

The following types of bias are more serious because they do not shrink with larger samples. They push results systematically in a particular direction regardless of sample size.

Response bias occurs when respondents give answers that do not reflect their true beliefs or behaviors. This often happens when the topic is sensitive or when the presence of an interviewer creates social pressure. A survey on hand-washing habits conducted by a hygiene inspector at a food preparation facility will likely overstate how often workers wash their hands, because workers know what answer the inspector wants to hear.

Nonresponse bias occurs when individuals selected for the sample cannot be reached or choose not to participate, and those non-respondents differ systematically from those who do respond. A survey mailed to parents about school satisfaction will be completed at higher rates by engaged parents than by disengaged ones. The result overstates satisfaction among the parent community as a whole.

Undercoverage bias occurs when part of the population is excluded from the sampling frame entirely. A survey about internet access conducted through online banner advertisements cannot reach people who lack internet access. Those excluded are precisely the group most relevant to the question.

Wording effect bias occurs when the phrasing of a question influences the answer. Consider the difference between these two ways of asking essentially the same thing:

- ”Do you support stricter emissions standards for vehicles?”
- ”Do you support stricter emissions standards for vehicles, even if it means higher car prices and fewer model choices?”

The second version frames the question with costs attached, making a ”no” response far more likely even among people who genuinely support the policy. Neutral, direct wording is essential for unbiased responses.

Exercise 5 Diagnosing Bias

Identify the most likely type of bias in each scenario and explain why it applies.

1. A streaming platform sends a satisfaction survey to all subscribers but receives responses from only 8% of them. The platform reports that 91% of subscribers are satisfied.
2. A survey on recycling habits is conducted by going door-to-door in affluent neighborhoods only, because those streets are easier and safer to walk.
3. An interviewer asks employees: ”Our new open-plan office is designed to encourage collaboration. How much has it improved your productivity?”
4. A survey on alcohol consumption is conducted at a community health fair by a team wearing hospital uniforms. Respondents significantly underreport their weekly intake.

Working Space

Answer on Page 22

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 10)

1. The population is all households in the city.
2. The frame is the city's water billing records.
3. The sample is the 200 households selected and interviewed.
4. No. A census would require interviewing every household in the city. Only 200 were selected here.

Answer to Exercise 2 (on page 11)

- **Library volunteers:** Census. With only 8 volunteers, asking all of them takes almost no effort.
- **Battery lifespan:** Sample. Testing can destroy the batteries, so a census would leave nothing to sell.
- **Commute times:** Sample. The population of 40 million is far too large for a full census to be practical.
- **Dietary restrictions:** Census. With only 22 guests, collecting complete information is straightforward and important for planning.

Answer to Exercise 3 (on page 13)

1. Number the 300 drivers from 001 to 300. Enter a random number table at any point and read three-digit numbers. Select any driver whose number falls between 001 and 300, skipping values from 301 to 999 and any numbers that have already been selected, until 25 drivers have been chosen. Alternatively, use `randInt(1,300,25)` on a TI-84, ignoring repeats.
2. This is a convenience sample. Drivers who are easy to reach at shift changes may work particular routes or shifts that differ systematically from others. For example,

they may have different safety exposure than drivers on overnight or rural routes.

3. No. An SRS requires that every possible group of 25 drivers from all 300 has an equal chance of being chosen. This procedure cannot produce a sample where all 25 come from the same depot, so not all groups of 25 are possible. This is a stratified sample, not an SRS.

Answer to Exercise 4 (on page 17)

- **Supermarket:** Cluster sampling. The stores are clusters, which means entire clusters are included once selected.
- **University:** Stratified random sampling. The four faculties are strata. An SRS is taken from each.
- **Water utility:** Systematic sampling. Random start followed by every 25th unit.
- **Film festival:** Simple random sample. Every possible group of 90 attendees is equally likely.

Answer to Exercise 5 (on page 20)

1. Nonresponse bias. The 92% who did not respond are unlikely to be satisfied in the same proportions as those who did. Highly satisfied and highly dissatisfied subscribers are both more motivated to respond than those with neutral feelings, but the pattern of non-response is unknown and the reported figure cannot be trusted.
2. Undercoverage bias. Residents of lower-income neighborhoods are excluded from the frame entirely. Recycling behavior may differ systematically by neighborhood type, so the results cannot generalize to the full population.
3. Wording effect bias. The question assumes the office has improved productivity and asks only "how much." Respondents are not given a neutral opportunity to say productivity has not changed or has declined.
4. Response bias. The formal appearance of the interviewers creates social pressure to give a health-conscious answer. Respondents adjust their reported behavior to match what they believe the interviewers expect.

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